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Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 23-03-2026

Responsive E-Commerce Interface with State-Persistent Cart

SDG: SDG 4 – Quality Education

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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Background of the problem:

Many small-scale e-commerce websites do not maintain cart data after page refresh or browser close. Traditional static websites lack dynamic cart management and persistent state handling.

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Why this project is important:

This project demonstrates how to build a responsive online shopping interface with cart persistence using modern frontend technologies.

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Real-world relevance:

Used in online shopping platforms, digital product stores, and retail web applications.

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Brief overview of the system:

A responsive e-commerce web application that allows users to browse products, add them to cart, and maintain cart state using Mysql WorkBench.

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PROBLEM STATEMENT



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In today's digital era, e-commerce platforms play a crucial role in online retail and digital marketplaces. However, many small-scale or beginner-level web applications lack proper state management and persistent cart functionality. In traditional static web applications, when users add products to their cart and refresh the page or close the browser, the cart data is lost. This results in a poor user experience and reduces customer satisfaction. Additionally, many simple e-commerce interfaces do not implement modular frontend architecture, dynamic product rendering, or real-time cart updates. Without proper storage mechanisms, the system fails to maintain user session data, selected products, and cart totals efficiently. This leads to inconsistency in cart behavior and makes the system unreliable for real-world use.

OBJECTIVES



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1. To implement cart functionality using MySQL database in MySQL Workbench persistent storage instead of LocalStorage.

2. To dynamically render products and categories.

3.To provide smooth user experience with real-time cart updates.

4.To design and develop a responsive e-commerce web interface using HTML, CSS, JavaScript, and Node.js.

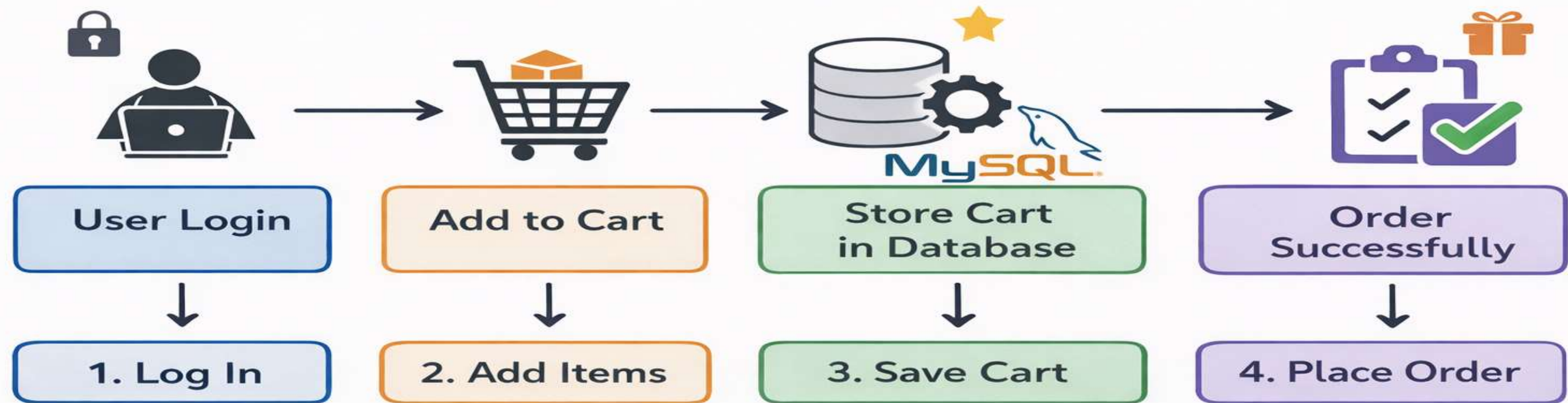
PROPOSED SYSTEM



1. The proposed system is a responsive e-commerce web application.
2. It allows users to browse products by categories.
3. Users can add and remove products from the cart.
4. Cart data is stored in MySQL database in MySQL Workbench .
5. The cart remains persistent even after page refresh.
6. Node.js server is used to serve application files.
7. The system reduces cart data loss issues.
8. It improves user experience and interface responsiveness

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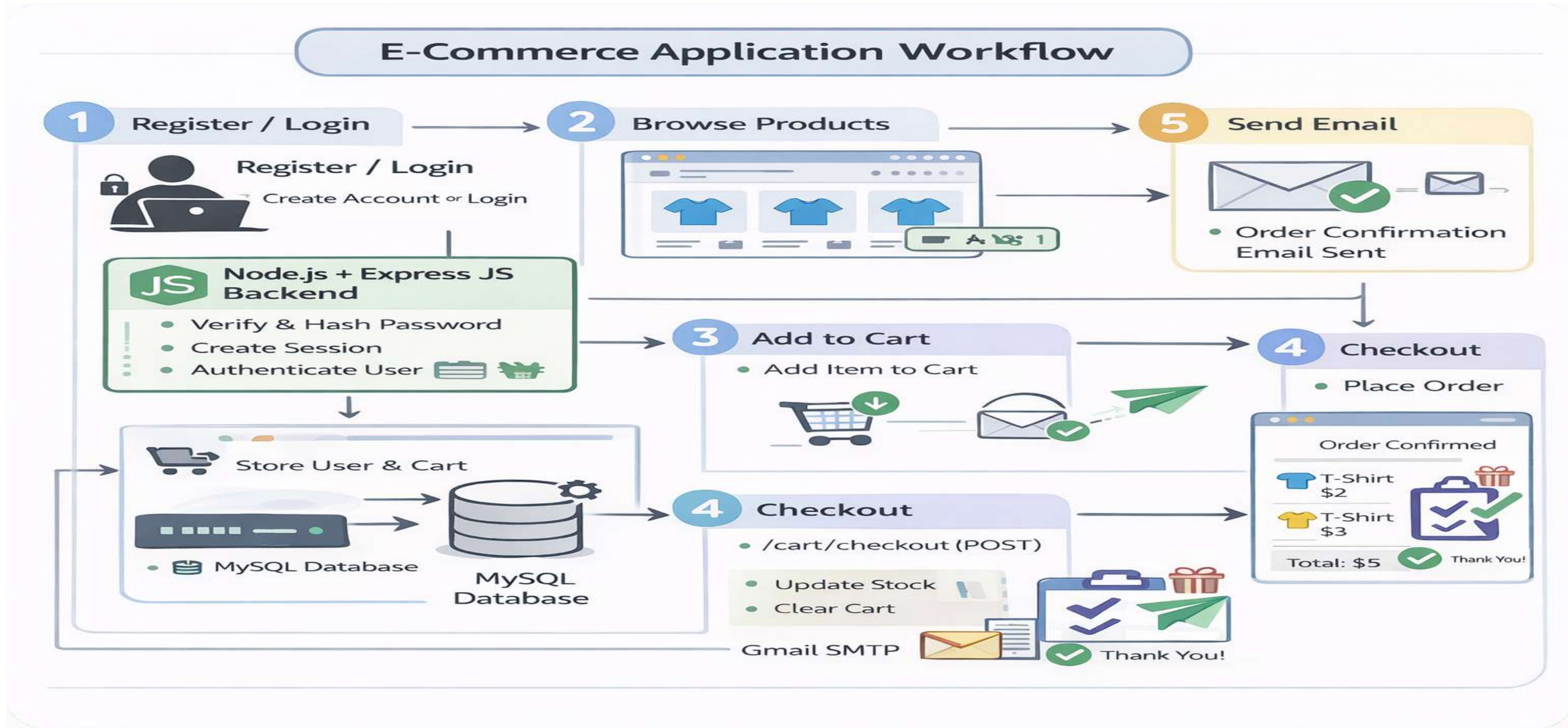
PROPOSED SYSTEM ARCHITECTURE



WORKFLOW



Dataflow diagram:



TECHNOLOGIES USED



Frontend:

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1.HTML5

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2.CSS3

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3.Javascript

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Backend:

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1.Node.js

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2.MySQL Workbench

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Tools:

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1.Visual Studio Code

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Software Requirements:

1.Windows

2. Web Browser

MODULE DESCRIPTION



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Module 1: Product Management:

- Display Products Dynamically.
- Organizes Products By Category.

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Module 2: Cart Management:

- Add To Cart
- Remove From Cart
- Update Quantity
- Calculate Total

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Module 3: State Persistence Module:

- Stores Cart In MySQL Workbench.
- Retrieves Data After Referesh.

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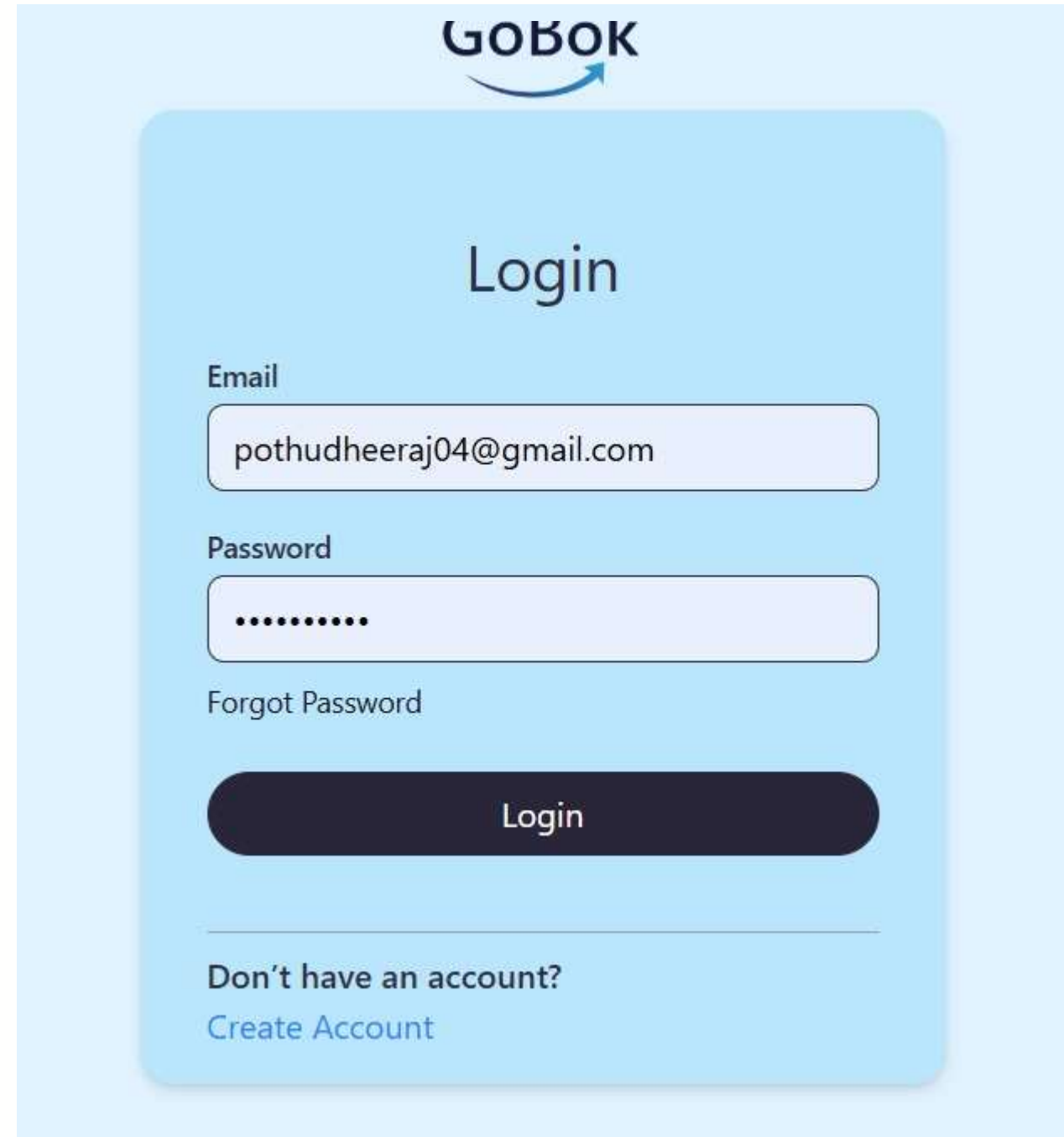
Module 4: Server Module:

- Node.js server to host application

IMPLEMENTATION SCREENSHOTS



Login Page :



The screenshot shows a login page for 'GOBOK'. At the top, the 'GOBOK' logo is displayed with a blue curved arrow underneath it. The word 'Login' is centered below the logo. There are two input fields: 'Email' with the value 'pothudheeraj04@gmail.com' and 'Password' with masked characters '.....'. Below the password field is a link for 'Forgot Password'. A dark blue 'Login' button is positioned below the form fields. At the bottom, there is a link that says 'Don't have an account? Create Account'.

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IMPLEMENTATION SCREENSHOTS



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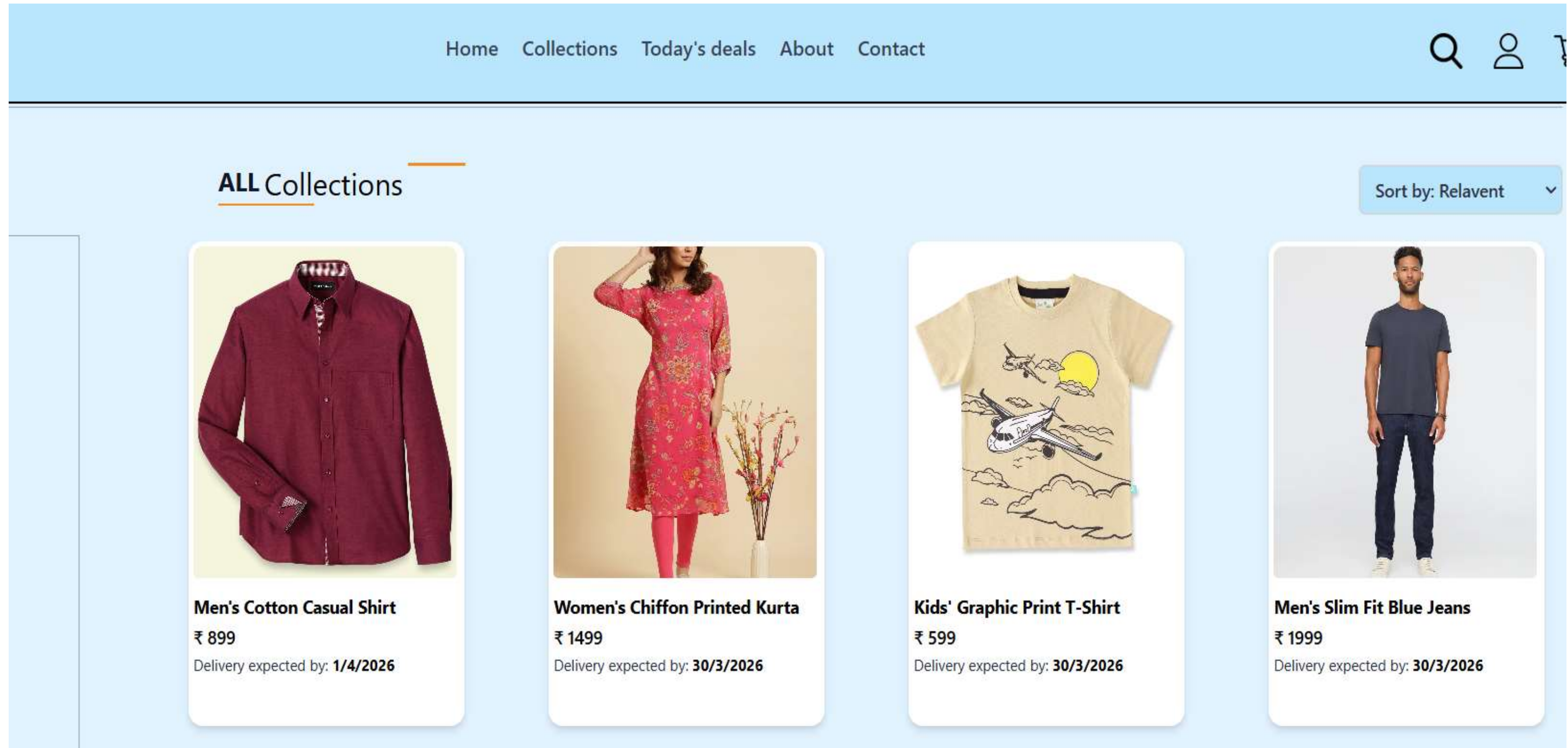
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IMPLEMENTATION SCREENSHOTS



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GoBok

HomeCollectionsToday's dealsAboutContact

Q

Your Cart Products



Tailored from premium cotton, this casual shirt ensures all-day comfort while maintaining a crisp look. Designed with a modern fit and button-down collar for versatility. Breathable fabric makes it suitable for summer wear. Pairs well with both jeans and trousers. Durable stitching ensures long life. Available in classic solid colors for timeless style.

₹899

M

Qty : 1

Remove



Kids' Graphic Print T-Shirt

Made from soft cotton, this graphic print t-shirt keeps kids comfortable during playtime. The vibrant design adds fun and energy to everyday outfits. Lightweight and breathable for all-day wear. Perfect for pairing with shorts or jeans. Durable stitching withstands rough use. Easy to wash and quick drying.

₹599

L

Qty : 2

Remove

Price Details

Price (5 items) :

₹ 5291.00

Protect Promise Fee

₹ 25.00

Total Amount

₹ 5316.00

Place Order

IMPLEMENTATION SCREENSHOTS



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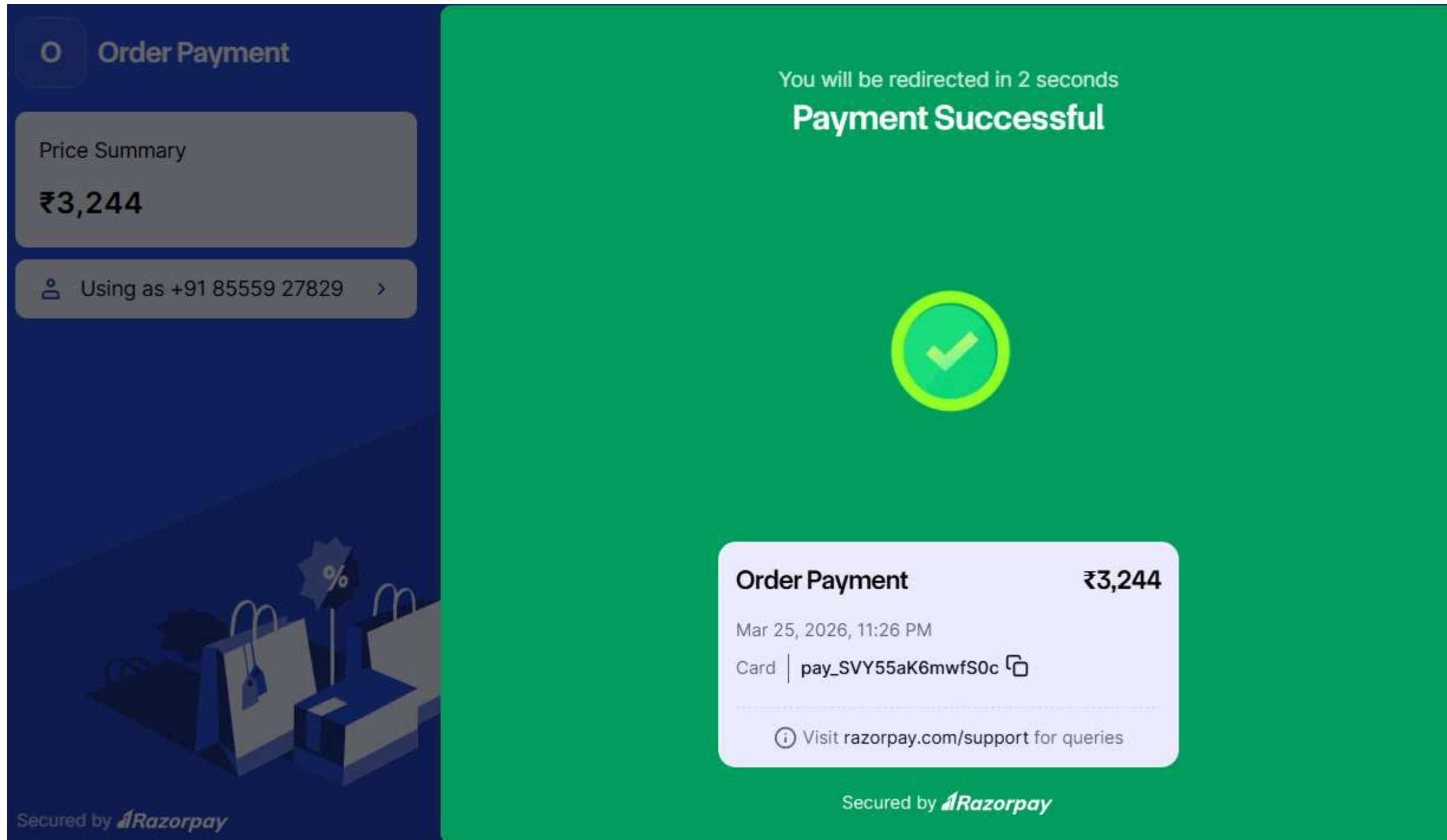
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ADVANTAGES & LIMITATIONS



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Advantages:

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1. State Persistence using Mysql Database.

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2. Responsive UI design.

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3. Faster cart operations.

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4. Lightweight application.

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Limitations:

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1. Limited Scalability.

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2. Weak Support for Advanced Features.

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3. Less Efficient for Complex Transactions.

4. Data Security Limitations.

Thank you

